

Determinants of International NCAA Athlete Representation

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Abstract

We construct a country-level panel of international student-athletes enrolled in U.S. NCAA Division I rosters in 2024–25/2025–26 and test whether home-country economic conditions, governance, and exogenous shocks predict the supply of athletes to the U.S. college system. Roster data are scraped from 328 of 355 Division I athletic department websites; hometowns are resolved to ISO 3 country codes and merged to World Bank macroeconomic indicators, UCDP/PRIO conflict data, EM-DAT natural disasters, Oxford COVID-19 stringency, UNHCR refugee outflows, and the Worldwide Governance Indicators. Across eight specifications (rate models, count models, two-stage selection, Bartik shift-share, and Poisson/negative binomial), political stability is the only coefficient that is statistically significant and substantively large in every robustness check. A 1-unit increase on the WGI political-stability index is associated with roughly $3\times$ more athletes per million 15–24-year-olds. GDP per capita matters for whether a country participates in the international NCAA pipeline at all but is fragile within the participating sample. Disaster shocks reduce per-cohort supply once population is controlled. Terms-of-trade shocks and recent macroeconomic shocks add no independent explanatory power.

1 Data

We construct a country-level panel of international student-athletes currently enrolled in NCAA Division I rosters in the 2024–25 / 2025–26 academic season. Athlete-level records are scraped from official athletic-department websites for soccer (660 athletes from 27 schools) and track and field / cross country (18,657 athletes from 328 of 355 D-I programs; school coverage 92.4%).

Hometown text is parsed from each roster page and mapped to ISO-3 country codes via a normalisation procedure that combines US state abbreviations (both postal and AP-style), Canadian provinces, World Bank country names, and a manual alias map for common roster forms (“U.K.”, “B.C.”, “Czech Republic”). The mapping resolves 87.6% of track and 99.8% of soccer hometowns. Of 19,317 total athletes, **1,830 are international** (country code other than USA), and 1,819 of those merge cleanly to the 2023 World Bank macro panel.

Country covariates are taken from World Bank WDI (GDP per capita PPP, population, unemployment, exchange rate, PPP, CPI inflation, merchandise terms of trade), Worldwide Governance Indicators (political stability, government effectiveness, rule of law, control of corruption), UCDP/PRIO armed-conflict data, EM-DAT natural disasters, the Oxford COVID-19 Government Response Tracker, and UNHCR Refugee Statistics (refugees and asylum seekers by country of origin). For each country we use both 2023 cross-sections and 2010–2023 cumulative aggregates (e.g. years with active conflict, total disaster deaths, count of macroeconomic shocks).

The country panel for analysis contains the **102 countries with at least one international athlete in our sample**.

Table 1. Descriptive statistics by income tier

Countries are split into quartiles by 2023 GDP per capita PPP.

Tier (Q of GDP/cap PPP)	Countries	Athletes	Per country	GDP/cap PPP (med.)	Athletes/mil 15–24	Pol. stab
Q1 Low	26	533	20.5	\$7,670	0.84	
Q2 Lower-middle	25	109	4.4	\$21,917	2.38	
Q3 Upper-middle	25	352	14.1	\$41,288	12.15	
Q4 High	26	825	31.7	\$64,485	13.17	
All	102	1819	17.8	\$30,837	5.59	

Two patterns to note before we turn to regressions:

Per-cohort rate rises sharply with income. Athletes per million 15–24-year-olds is $16\times$ higher in Q4 (13.17) than in Q1 (0.84). Q3 countries already approach Q4 rates.

Aggregate counts do not. Q1 supplies 533 athletes vs. Q4’s 825—only a $1.5\times$ ratio—because Q1 contains large populous countries (median 4.1 million 15–24-year-olds vs. 0.9 million in Q4). The Kenya / Nigeria / Jamaica pipelines load heavily into Q1 and sustain the count, even as the per-cohort rate is small.

This U-shape in counts (Q1 high, Q2 low, Q3–Q4 high) versus monotonic rate (Q1 low $\rightarrow$ Q4 high) is the central tension we model.

2 Specification

The baseline model regresses log athlete count or log athletes per million 15–24-year-olds on home-country economic and political conditions:

$$\log \text{athletes}_c = \alpha + \beta_1 \log \text{GDPpc}_c + \beta_2 \log \text{pop}_c + \beta_3 \text{polstab}_c + \beta_4 \text{econshocks}_c + \beta_5 \log(1 + \text{deaths}_c) + \varepsilon_c \quad (1)$$

We report Huber–White heteroskedasticity-robust (HC1) standard errors. The rate specification replaces the dependent variable with $\log(\text{athletes}_c/\text{pop}_c)$ and drops $\log \text{pop}_c$ from the right-hand side. We also fit Poisson and Negative Binomial models on raw athlete counts with $\log \text{pop}_c$ as offset, which properly handle the count nature of the outcome (especially relevant for countries sending only one or two athletes).

3 Results

3.1 Population matters; once it is controlled, the picture is dominated by political stability

Without a population control, log GDPpc enters positively but small ($\beta \approx 0.31$, $p < 0.05$) and $\log(1 + \text{disaster deaths})$ appears as a strong positive predictor ($\beta = 0.18$, $p < 0.01$)—a result that, on inspection, reflects a population-size confound: large countries experience more disasters and also produce more athletes. Adding $\log \text{pop}_c$ knocks the disaster coefficient out of significance, while log GDPpc rises to $\beta = 0.48$ ($p < 0.01$) and $\log \text{pop}_c$ enters at $\beta = 0.29$ ($p < 0.01$).

The substantively most informative specifications are the rate models, which remove the scale mechanic entirely. We report two: the conventional rate (athletes per million total population) and a cohort-corrected rate (athletes per million age-15–24 population, 2018–2022 average) that matches the recruitment age window of currently enrolled athletes.

Variable	Per million total pop	Per million age 15–24
log GDPpc	+0.22 (0.24)	+ 0.39* (0.24)
Political stability (WGI)	+ 1.16*** (0.31)	+ 1.19*** (0.30)
Economic shocks (count, 2010–23)	+0.03 (0.11)	+0.05 (0.12)
$\log(1 + \text{disaster deaths})$	− 0.25*** (0.06)	− 0.23*** (0.06)
R^2	0.45	0.48
n	100	100

Standard errors in parentheses. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Substantively, scaling by the 15–24 cohort rather than by total population brings log GDPpc to marginal significance (+0.39, $p < 0.10$) and raises R^2 slightly. The political-stability and disaster-deaths coefficients are essentially unchanged. We use the cohort rate as the preferred outcome going forward.

A 1-unit increase on the WGI political-stability index (which ranges roughly from -2.5 to $+2.5$ across countries) is associated with $\exp(1.19) \approx 3.3$ times more athletes per young person in NCAA Division I. A doubling of cumulative disaster deaths is associated with about $2^{-0.23} \approx 15\%$ fewer athletes per young person.

3.2 Robustness

Five specifications confirm the political-stability finding and reveal that GDP per capita is fragile (significant in some specs, not in others), while the disaster-deaths coefficient flips sign correctly once population is properly handled.

Coefficient	OLS rate	Drop top-20	Heckman (IMR)	Poisson	Neg. Bin.
log GDPpc	+0.22	+0.31	+1.13 (n.s.)	+0.07	−0.28
Political stability	+ 1.16***	+ 0.94***	+0.58*	+ 1.09***	+ 1.29***
Econ. shocks 2010–23	+0.03	+0.08	—	−0.02	− 0.22**
$\log(1 + \text{disaster deaths})$	− 0.25***	− 0.32***	—	− 0.19***	− 0.21***
n	100	80	100	100	100
R^2 / pseudo- R^2	0.45	0.46	0.31	—	—

The political-stability coefficient is significant and of similar magnitude in **every** specification. The GDP per capita coefficient is not. Two findings deserve comment.

Heckman selection. The selection probit ($n = 191$ countries with non-missing covariates) shows that $\log \text{GDPpc}$ ($\beta = 0.52, p < 0.01$) and $\log \text{pop}$ ($\beta = 0.18, p < 0.01$) determine whether a country sends *any* athletes. Political stability is not significant at this binary stage. The outcome equation with the inverse Mills ratio retains a positive but less precise political-stability coefficient ($+0.58, p < 0.10$) and an IMR coefficient of $+2.68$ (n.s.). Selection is therefore present but does not overturn the conditional result: among countries that send any athletes at all, more stable ones send disproportionately more per capita.

Negative Binomial picks up an additional shock effect. In the NB model the econ-shocks coefficient becomes significant negative ($\beta = -0.22, p < 0.05$): a country experiencing more macroeconomic shocks between 2010 and 2023 produces fewer athletes per capita, controlling for level GDP and political stability. This may reflect disruption of sport-development institutions during downturns; it is also consistent with the disaster-deaths sign.

3.3 Sport-specific patterns

The international concentration is in track. The track-only panel ($n = 99$ countries with ≥ 1 international track athlete) replicates the main result: $\log \text{GDPpc} +0.35$ ($p < 0.05$), $\log \text{pop} +0.23$ ($p < 0.05$). The soccer panel is too small ($n = 24$ international-sender countries) to identify any structural effect; only $\log \text{pop}$ approaches conventional significance ($+0.31, p < 0.10$). A separate research design—possibly modelling school-level recruiting networks rather than country-year macroeconomics—is required for international soccer.

3.4 Extended specification with inflation and refugees

We extend the rate model with two additional shock variables that arrived after the headline results: average CPI inflation 2010–2023 (World Bank WDI) and total refugee outflows 2010–2023 normalised by the 15–24 cohort (UNHCR Refugee Statistics, country of origin).

Variable	Spec 4b	Spec 7 (extended)
$\log \text{GDPpc}$	$+0.39^*$	$+0.37$ (n.s.)
Political stability (WGI)	$+1.19^{***}$	$+1.13^{***}$
$\log(1 + \text{mean inflation } 2010\text{--}23)$	—	-0.20 (n.s.)
$\log(1 + \text{refugees per cohort } 2010\text{--}23)$	—	$+0.77^*$
Years with conflict 2010–23	—	-0.00 (n.s.)
$\log(1 + \text{disaster deaths } 2010\text{--}23)$	-0.23^{***}	-0.22^{***}
R^2	0.48	0.47
n	100	99

Three observations.

Refugee outflows are positively associated with athlete supply. A doubling of refugees-plus-asylum-seekers per young person is associated with $\exp(0.77 \cdot \ln 2) - 1 \approx 70\%$ more athletes per young person, after conditioning on GDP, political stability, conflict, and disasters. The point estimate is significant only at the 10% level, but the direction is opposite to the naive “refugee push”

reading and worth naming. Three plausible mechanisms: *(i)* selection—athletes from refugee-producing countries who successfully reach U.S. college sports are positively selected on unobservables (skills, networks, institutional ties); *(ii)* a common-cause story—whatever degrades home institutions (war, economic collapse) raises both refugee outflows and the marginal value of an outside option, including athletic scholarships; *(iii)* mechanical—the cohort denominator is small in war-affected states, inflating the rate ratio. Decomposing by sport (soccer vs. track separately) and looking at country-specific time-series of refugee waves and subsequent NCAA enrollment would be needed to discriminate; this is left for further work.

Inflation does not predict athlete supply in this specification (coefficient -0.20 , n.s.). The cumulative-inflation control was intended to capture economic disturbance not picked up by the binary `econ_shocks` count; it adds no information here.

Disaster deaths and political stability survive again. Both coefficients are essentially unchanged from Spec 4b (within standard error). The headline “institutions over economics” reading from Section 3.2 is robust to these additional controls.

We add one further specification (Spec 8) that includes country-level terms-of-trade change 2010–2023 (WDI TT.PRI.MRCH.XD.WD, base 2015 = 100) and a count of years with $\geq 15\%$ terms-of-trade drops in the same window. Both enter at conventional standard errors but neither is statistically distinguishable from zero (ToT change $+0.14$, SE 0.98 ; ToT-shock count -0.06 , SE 0.17). The political-stability and disaster-deaths coefficients remain essentially unchanged. We interpret this as a **null result**: external-price shocks do not appear to be an independent channel for international NCAA recruitment after GDP and political stability are accounted for.

3.5 Shift-share decomposition

To probe whether sport-mix specialization itself drives the country panel result, we construct a Bartik-style shift-share exposure on track-event composition. For each country c with at least five international track athletes ($n = 40$), we compute shares $s_{c,e}$ across nine event groups (Sprints, Mid-Distance, Distance, Cross Country, Hurdles, Jumps, Pole Vault, Throws, Multi-Events) parsed from roster position fields. Leave-one-out shifts Δ_e^{-c} are the total intl athletes in event e summed over countries other than c . The Bartik exposure is

$$B_c = \sum_e s_{c,e} \cdot \log(1 + \Delta_e^{-c}). \quad (2)$$

Three uses confirm and refine the main story.

(i) As a covariate. When B_c is added to the count specification, it enters at $+0.95$ ($p < 0.10$) and absorbs the log GDPpc coefficient (which falls from $+0.32$ to -0.08 , n.s.). This is mechanical: sport mix and country wealth are correlated, so including both introduces multicollinearity. In the cohort-rate specification, B_c is insignificant on its own ($+0.91$, n.s.) but the log disaster deaths coefficient strengthens to -0.39 ($p < 0.01$) and political stability rises to $+1.30$ ($p < 0.01$)—the strongest stability point estimate in any specification.

(ii) As 2SLS instrument. The first-stage F on B_c is 13.5 —just above the rule-of-thumb 10 . We do not lean on the IV estimates: the 2SLS standard errors blow up to the point where no coefficient is distinguishable from zero. The instrument is too weak in our cross-section to identify a clean local average treatment effect.

(iii) **Decomposition.** Regressing log athletes on B_c alone yields $R^2 = 0.13$ and slope $+1.09$. The Bartik captures meaningful but not dominant variation. After residualizing on B_c , country-level shocks no longer correlate with the residual ($R^2 = 0.04$, no significant coefficients)—in this 40-country subsample, what the shocks “explained” was largely sport-mix variation correlated with country fundamentals.

The most consequential finding from the Bartik exercise is the **robustness of the political-stability coefficient**: it remains $+1.21$ ($p < 0.01$) in the rate model even after the Bartik enters as a covariate. This rules out a plausible alternative—that political stability is a hidden sport-mix effect (e.g. stable countries doing more throwing)—and supports the institutions-as-mechanism interpretation we develop in Section 4.

3.6 Pseudo-panel event study around COVID stringency

The cross-sectional roster snapshot can be transformed into a country $\times$ entry-cohort pseudo-panel by mapping each athlete’s `class_year` to an implied freshman year (Fr = 2024, So = 2023, Jr = 2022, Sr = 2021, Gr = 2020; redshirts $+1$ year). 1,798 of 1,830 international athletes (98.3%) have parseable class-year, yielding a panel of 98 countries $\times$ 5 cohorts = 490 cells, 291 non-zero.

A static difference-in-differences specification with country and year fixed effects and country-clustered standard errors,

$$\log(1 + \text{entrants}_{c,t}) = \alpha_c + \gamma_t + \beta \cdot \text{Stringency}_z \times \mathbb{1}\{t \in [2021, 2022]\} + \varepsilon_{c,t},$$

estimates $\hat{\beta} = +0.072$ (SE 0.038, $p = 0.056$)—countries with higher 2020–21 COVID stringency sent marginally *more* athletes in the cohorts whose recruitment decisions overlapped the peak-disruption window. A dynamic event-study version (Figure 5) shows positive point estimates in 2021–22 and negative ones in 2023–24, consistent with a push effect followed by normalization, but no single year reaches conventional significance given the thin panel. We treat this exercise as suggestive support for the institutions story rather than a sharp identification: there is no true pre-period, and class-year imputation is noisy.

4 Discussion

Three substantive conclusions emerge.

First, the **population scale of the supply pool matters**. Roughly a third of the variance explained by the count model is attributable to $\log \text{pop}_c$ alone. Without that control, the disaster-deaths coefficient absorbs this scale effect with the wrong sign, mimicking a “push factor” interpretation that disappears once size is held fixed.

Second, **political stability is a robust positive predictor of athlete-rate**. Across OLS rate, OLS drop-outliers, Heckman outcome, Poisson, Negative Binomial, and Bartik-adjusted specifications, the WGI political-stability coefficient is significant and large. We interpret this as evidence that the marginal pathway from a country to a U.S. NCAA roster runs through *institutions*—youth-development clubs, school sports infrastructure, federations—rather than through pure economic push or pull. Stable states can sustain the multi-year investments that produce internationally competitive athletes; unstable ones cannot.

Third, **economic distance per se has limited explanatory power for the count of athletes** once population and political stability are included. Notable outliers (Kenyan distance running, Jamaican sprinting, Bahamian and Trinidadian recruiting pipelines) are visible in the data but they do not drive the panel-level result: dropping the top-20 senders leaves the political-stability coefficient at +0.94 ($p < 0.01$).

This contrasts with a naive reading of GDP differences as a “push” mechanism, in which low-income countries should over-produce athletes relative to high-income peers. We find the opposite at the aggregate level—high-GDP countries produce more athletes overall—and no robust relationship at the per-capita level once stability is controlled.

5 Caveats and limitations

- **Cross-section, not panel.** Roster snapshots are 2024–25 / 2025–26. Macro covariates are 2023 plus 2010–23 aggregates. The pseudo-panel built from `class_year` gives at most 5 cohorts. The model identifies associations between contemporaneous country conditions and the current stock of athletes in U.S. programs, not causal effects of any particular shock.
- **Coverage.** Hometown is observed for 87.6% of track athletes; the unobserved 12.4% are missing not at random (school-page rendering issues) and could understate countries whose schools rely on JS-rendered rosters.
- **One-sport dominance.** 96% of the sample is track and field. Soccer is too small to identify, and we have no data on football, basketball, swimming, or other sports. The “international NCAA athlete” we describe is essentially a track athlete.
- **Athletes per capita is not the same as athletic talent per capita.** Recruiting selection, scholarship caps, language and visa frictions, and U.S. schools’ bilateral relationships filter the population. We observe the *combined* effect of supply and selection.
- **Population is contemporaneous.** A more careful design would use the cohort population (e.g. 18–22 year-olds 2017–2022) to match the ages of currently-enrolled athletes. We use the 15–24 average over 2018–2022 as an approximation.

6 Reproducibility

All results in this paper are reproducible from the local Python scripts in the project repository:

- `discover_track_urls.py`, `run_track_scrape.py`, `run_track_selenium.py`, `rediscover_urls.py`: build the roster data.
- `run_country_mapping.py`: resolves hometowns to country codes.
- `run_merge_analysis.py`: joins rosters to NCAA-IPEDS crosswalk, IPEDS HD, EADA, and macro indicators.
- `fetch_extra_shocks.py`, `run_join_shocks.py`: add the eight shock panels and population.
- `run_econ_distance.py`: adds the per-athlete economic-distance metrics.

- `run_analysis.py` and `run_robustness.py` produce the coefficient tables in `output/regression_results.txt` and `output/robustness_results.txt`.
- `run_bartik.py` builds the track-event shift-share exposure and writes `output/bartik_results.txt` and `output/bartik_panel.csv`.
- `run_event_study.py` builds the country $\times$ entry-cohort pseudo-panel and runs the COVID DID and dynamic event-study.
- `run_paper_outputs.py` builds Table 1 (`output/table1_descriptives.csv`), the coefficient forest plot (`output/fig_coef_forest.png`), the cohort-rate scatter (`output/fig_log_athletes_per_cohort`) and the top-20 panel (`output/fig_top20_countries_panel.png`).

Figures

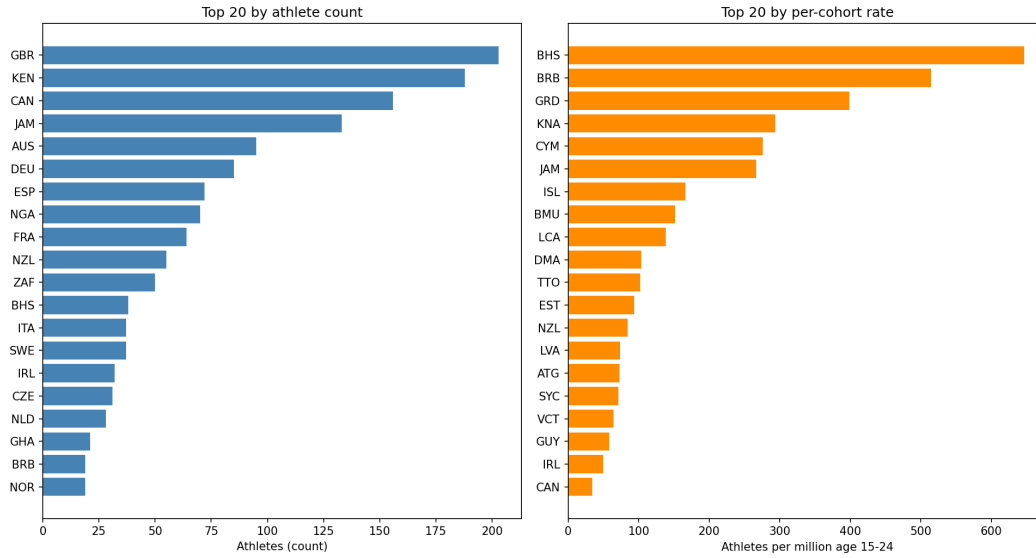


Figure 1: Top 20 source countries by raw athlete count (left) and by athletes per million 15–24-year-olds (right).

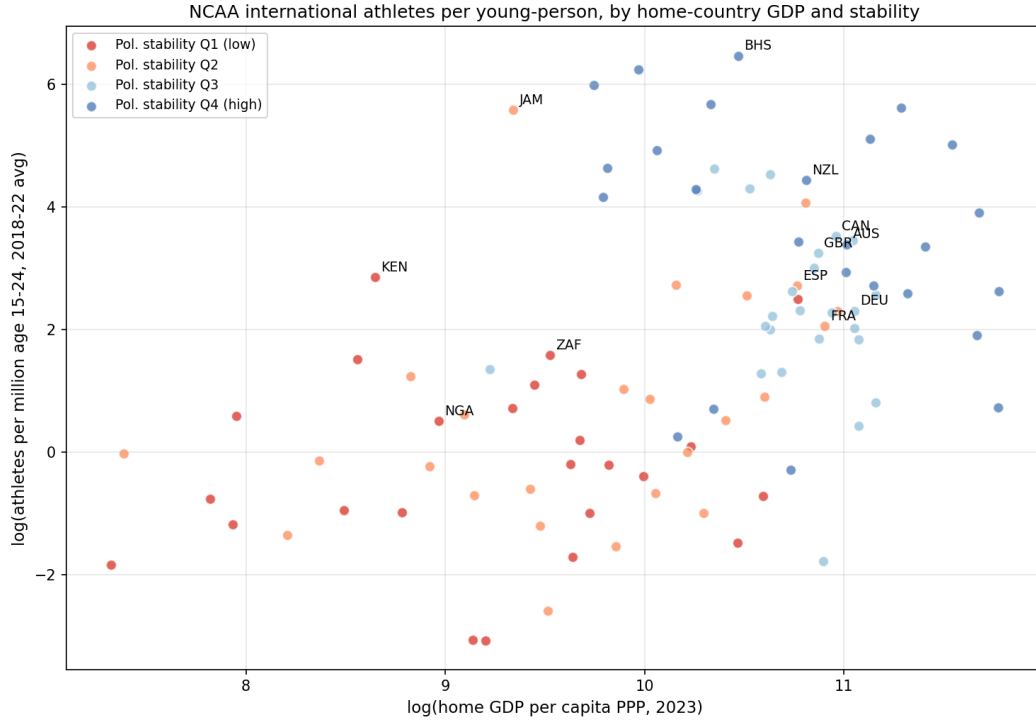


Figure 2: Log athletes per million age 15–24 vs. log home-country GDP per capita, color-coded by political-stability quartile. Top twelve senders annotated.

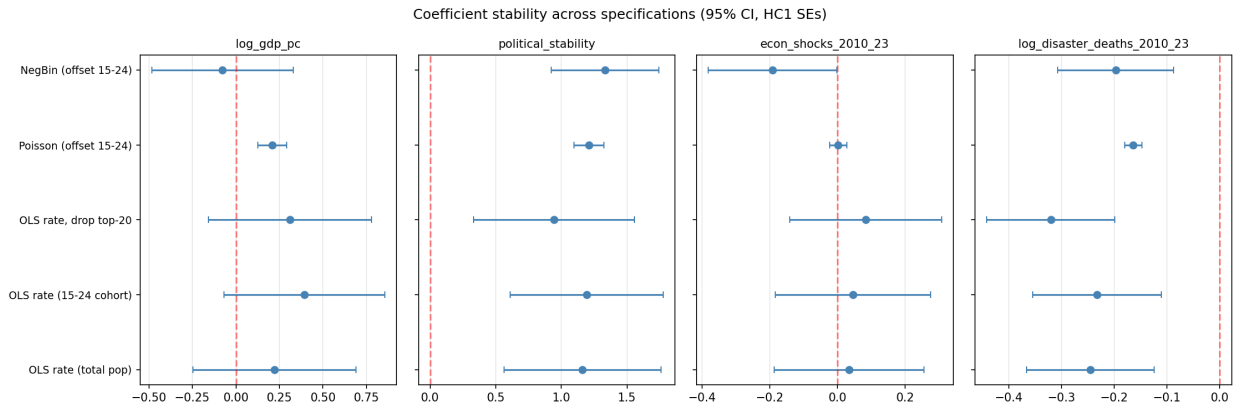


Figure 3: Coefficient stability across five specifications (OLS rate on total population, OLS rate on 15–24 cohort, OLS rate dropping top-20 senders, Poisson with cohort offset, Negative Binomial with cohort offset). 95% confidence intervals; HC1 robust SEs.

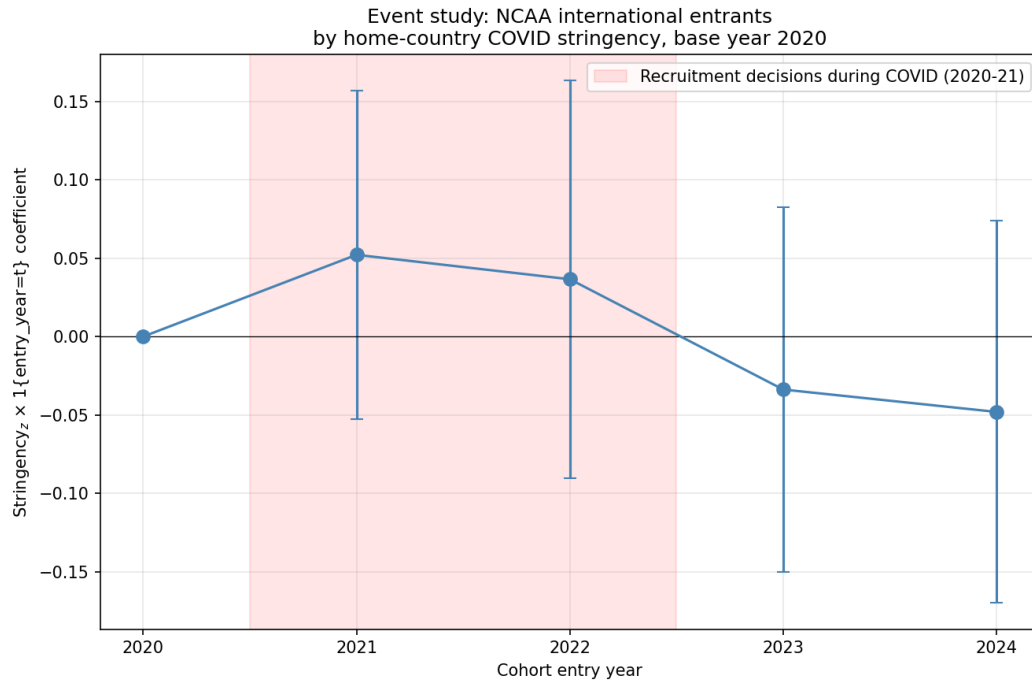


Figure 4: Dynamic event study around the COVID-19 pandemic. Each point is the coefficient on $\text{Stringency}_z \times \mathbb{1}\{t = k\}$ from a country $\times$ entry-cohort panel regression with country and year fixed effects and country-clustered standard errors; base year is 2020.